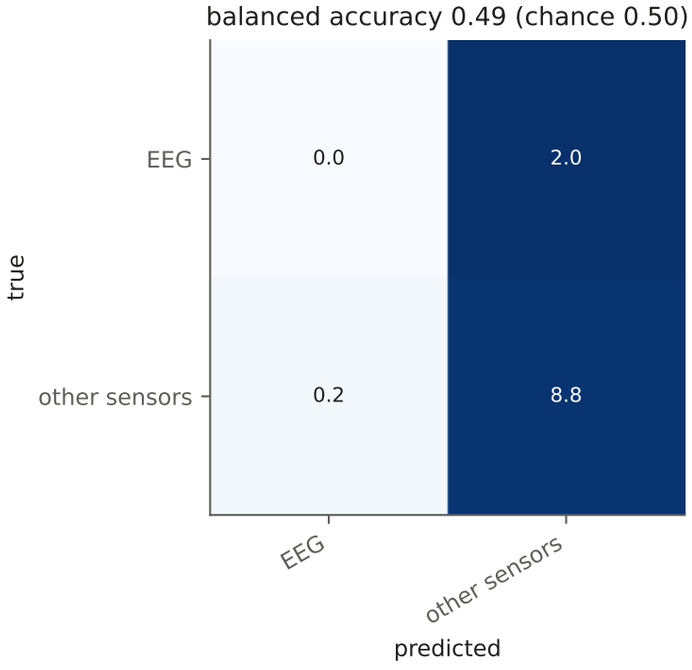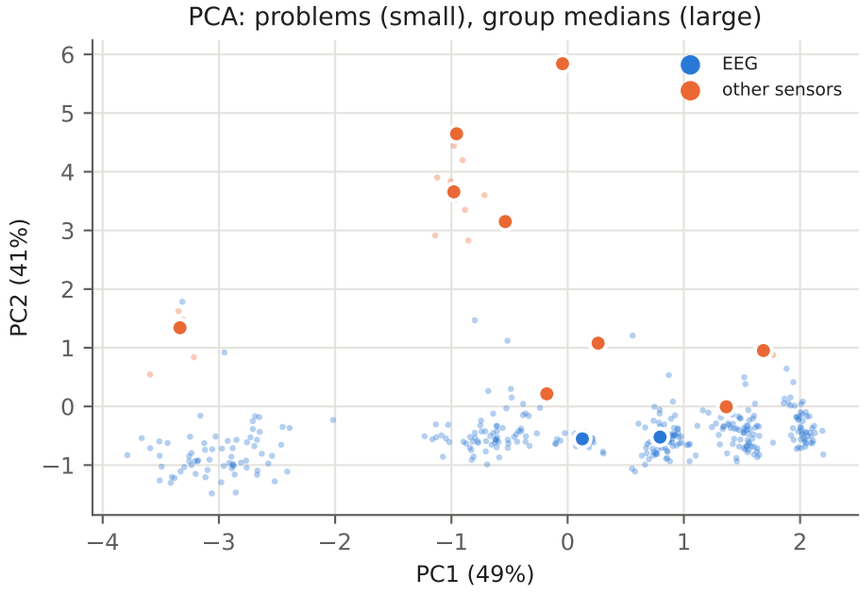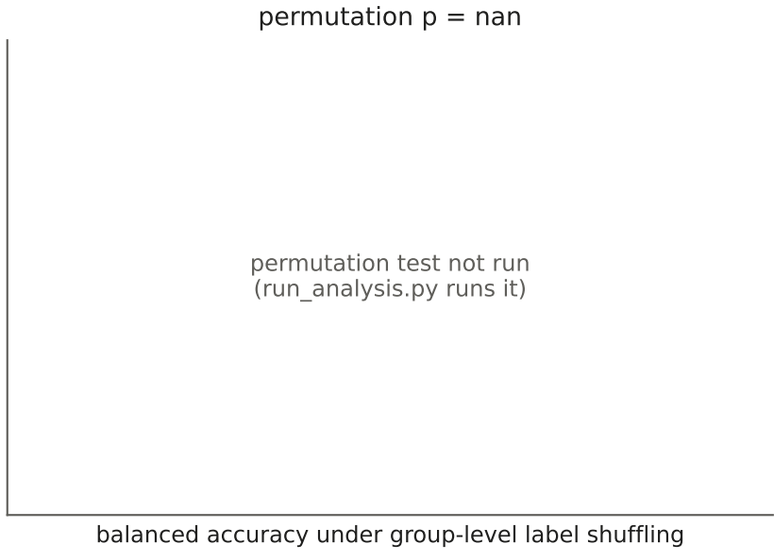q3b_covariance